# Human Activity Classification Using Sensor Data

*Machine Learning Mini-Project | Technical Write-up*

## 1. Problem Statement and Objectives

Human activity recognition uses wearable sensor measurements to infer physical activities. This project treats activity recognition as a supervised classification task using the PAMAP2 Physical Activity Monitoring dataset. The objectives are to explore and preprocess sensor data, compare two feature configurations, train six machine-learning classifiers, and evaluate performance using accuracy, precision, recall, F1-score, and confusion matrices.

## 2. Dataset Description

The workflow uses the nine PAMAP2 protocol files, subject101.dat through subject109.dat. Each subject file is loaded as 54 columns in the current notebook representation. The data include activity labels, heart rate, and inertial sensor readings from the hand, chest, and ankle. Activity ID 0 is removed, leaving 12 activity classes. The original dataset is not included in the repository.

Download PAMAP2 from the UCI Machine Learning Repository and place the protocol files under data/PAMAP2/Protocol/ before running the notebooks: https://archive.ics.uci.edu/dataset/231/pamap2+physical+activity+monitoring

## 3. Preprocessing and Feature Configurations

Data from the nine subjects are combined while retaining a subject identifier. Missing heart-rate values are interpolated separately within each subject. Rows that still contain missing values are excluded from the complete feature matrices. Two configurations are evaluated: Full features (52 features, including hand, chest, and ankle sensor measurements plus heart rate) and Limited features (18 features, using hand sensor measurements plus heart rate). The complete matrices contain approximately 1.92 million full-feature rows and 1.93 million limited-feature rows before splitting.

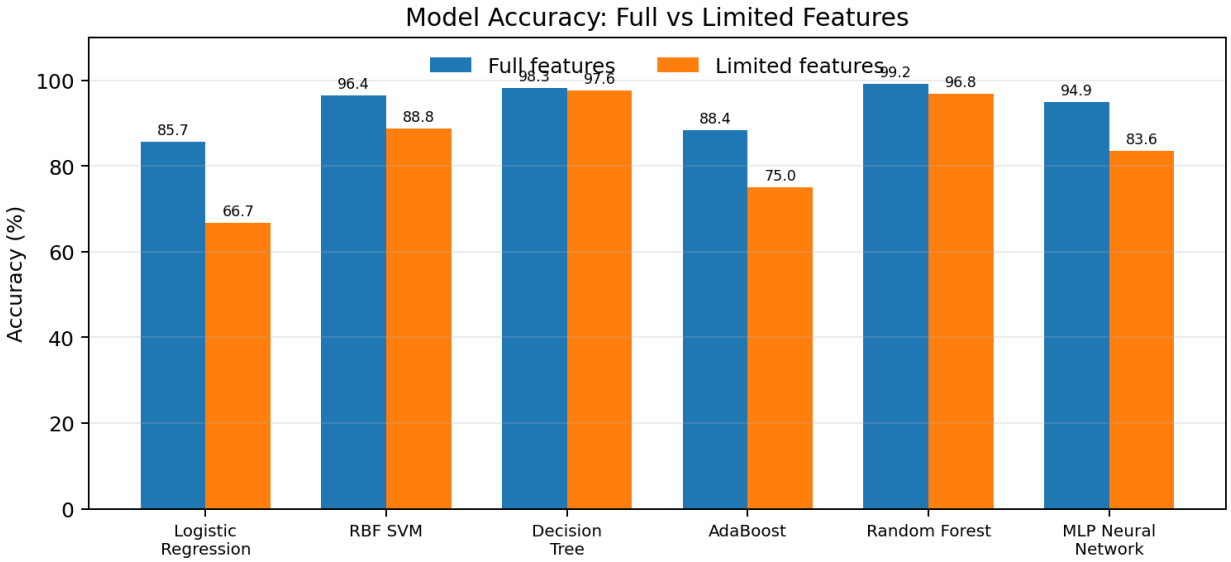
## 4. Models and Evaluation

The classifiers are Logistic Regression, Support Vector Machine with an RBF kernel, Decision Tree, AdaBoost, Random Forest, and Multi-Layer Perceptron (MLP) Neural Network. The main split uses 85% training and 15% testing, random_state=42, and stratification. Standardization is used for models requiring scaled inputs. The SVM experiment uses a separate sampled subset of 10,000 training and 20,000 testing examples per feature configuration due to computational cost.

# Results and Performance Comparison

## 5. Accuracy Results

The chart compares the reported test accuracies for the six classifiers. The SVM uses a separate sampled test subset, so its result should be interpreted with that evaluation difference in mind.



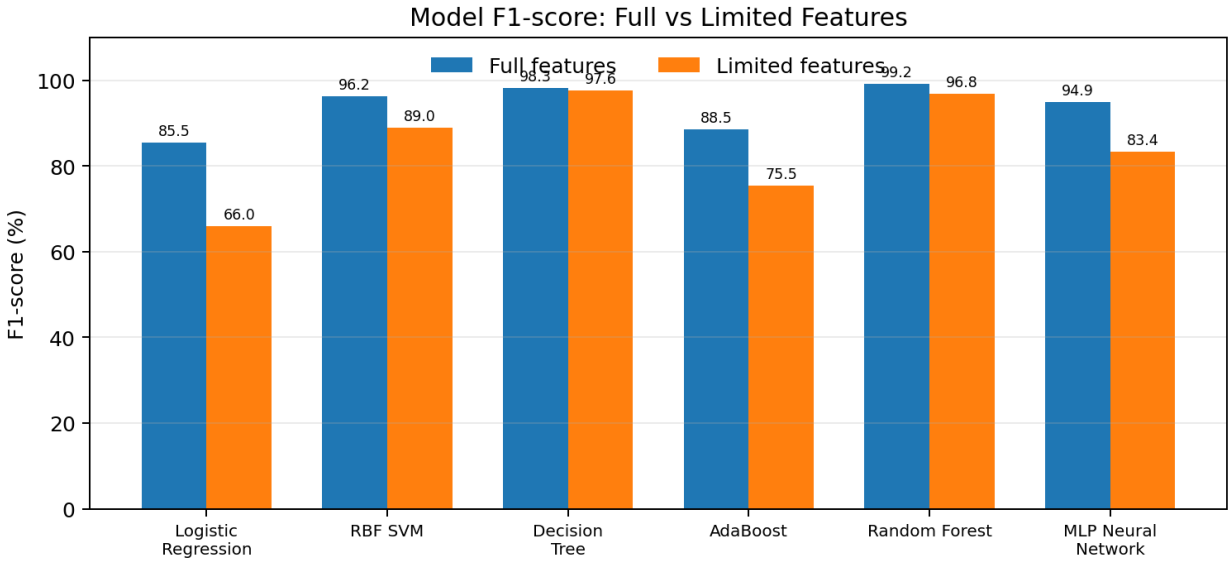

## Accuracy values (%)

| Model | Full features | Limited features |
|---|---|---|
| Logistic Regression | 85.70 | 66.72 |
| RBF SVM | 96.44 | 88.82 |
| Decision Tree | 98.31 | 97.61 |
| AdaBoost | 88.40 | 75.02 |
| Random Forest | 99.22 | 96.81 |
| MLP Neural Network | 94.92 | 83.59 |

# Discussion and Conclusion

## 6. F1-score Comparison

F1-score combines precision and recall and provides a balanced summary of classification performance. The chart below compares the reported F1-scores for the full and limited feature configurations.




## 7. Discussion

Random Forest achieved the highest reported accuracy with full features (99.22%), followed by Decision Tree (98.31%). With limited features, Decision Tree achieved the highest reported accuracy (97.61%), followed by Random Forest (96.81%). All six classifiers had lower reported accuracy with limited features; the decrease was comparatively small for the two tree-based models. This suggests that the impact of feature reduction depends on the classifier.

The results directory contains the performance CSV, comparison plots, and confusion matrices. Confusion matrices can be used to inspect which activity classes are most frequently confused.

## 8. Limitations

The main evaluation uses a row-wise random split rather than holding out entire subjects. Therefore, the results do not establish performance on entirely unseen individuals. The SVM uses a separate sampled test set, so its accuracy is not strictly equivalent to the other models' main test scores. Results may also vary with preprocessing choices, data distribution, and sensor devices.

## 9. Conclusion

This project demonstrates a notebook-based workflow for wearable-sensor activity classification, including exploration, preprocessing, feature construction, model training, quantitative evaluation, and visualization. Random Forest performed best with full features, while Decision Tree performed best with limited features. Future work could evaluate subject-independent splits to better estimate performance on new individuals.

## Reference

Reiss, A. and Stricker, D. (2012). Introducing a New Benchmarked Dataset for Activity Monitoring. PAMAP2 Physical Activity Monitoring dataset, UCI Machine Learning Repository: https://archive.ics.uci.edu/dataset/231/pamap2+physical+activity+monitoring